SMART EXPLAINABLE AI MODEL FOR PERSONALIZED DISEASE RISK ASSESSMENT AND EARLY DISEASE PREDICTION

A Project Report submitted for the partial fulfillment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

SUBMITTED BY

HEMANTH KUMAR PATTEM
ROLL NO: 24691F0071

Under the Guidance of

Dr. SRINIVASAN JAGANNATHAN, M.C.A., Ph.D.

Assistant Professor

Department of Computer Applications

DEPARTMENT OF COMPUTER APPLICATIONS

MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE

UGC AUTONOMOUS

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DEPARTMENT OF COMPUTER APPLICATIONS

BONAFIDE CERTIFICATE

This is to certify that the project work entitled **"SMART EXPLAINABLE AI MODEL FOR PERSONALIZED DISEASE RISK ASSESSMENT AND EARLY DISEASE PREDICTION"** is a Bonafide work carried out by Hemanth Kumar Pattem (Reg. No. 24691F0071), submitted in the partial fulfillment of the requirements for the award of the degree of Master of Computer Applications in Madanapalle Institute of Technology & Science, Madanapalle, affiliated to Jawaharlal Nehru Technological University Anantapur, Ananthapuramu the academic year 2025-2026.

PROJECT GUIDE	**HEAD OF THE DEPARTMENT**
Dr. J. SRINIVASAN JAGANNATHAN, M.C.A., Ph.D.	**Dr. N. NaveenKumar, M.C.A., Ph.D.**
Assistant Professor	Professor & Head
Department of Computer Applications	Department of Computer Applications

Submitted for viva voice examination held on ____________________

Internal Examiner	**External Examiner**
Date:	Date:

DECLARATION

I, Hemanth Kumar Pattem **(Reg. No. 24691F0071)** hereby declare that the project entitled **“SMART EXPLAINABLE AI MODEL FOR PERSONALIZED DISEASE RISK ASSESSMENT AND EARLY DISEASE PREDICTION”** is done by me under the guidance of **Dr. J. SRINIVASAN JAGANNATHAN, M.C.A., Ph.D.,** submitted in partial fulfillment of the requirements for the award of degree of Master of Computer Applications at **MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE**, Madanapalle affiliated to Jawaharlal Nehru Technological University Anantapur, Anantapuram during the academic year 2025-2026. This work has not been submitted by anybody towards the award of any degree.

Date:

Place: Madanapalle

Signature of the Student

(Hemanth Kumar Pattem)

(Roll No.: 24691F0071)

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Hemanth Kumar Pattem

ABSTRACT

The utilization of artificial intelligence (AI) in the healthcare industry allows for tailored risk assessment and proactive disease identification. However, most systems today deal with the forecasting of a singular disease, rely on particular clinical variables, and lack explainability. Using the Pima Indians Diabetes, Cleveland Heart Disease, and Wisconsin Breast Cancer Datasets, this study introduces the Adaptive Explainable AI Framework. Diabetes, heart disease, and breast cancer risk prediction is most by the explained risk and diagnosed including the structured Electronic Health Records (EHR).

To improve prediction accuracy, robustness, and generalizability for the different disease profiles, a hybrid stacking ensemble is constructed by combining Random Forest, XGBoost, and Logistic Regression as base learners and a Logistic Regression meta-learner. To achieve transparency and trust in the healthcare model, XAI (Explanable AI) components SHAP for global feature importance and LIME for individual explanation are integrated. These focus attention on the critical risk factors of breast cancer such as area_worst, concave points_worst and concave points_mean along with diabetes including glucose, BMI and age.

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LIST OF ABBREVAITIONS

S.NO.	ABBREVIATED TERM	EXPANSION
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	XAI	Explainable Artificial Intelligence
4	SHAP	SHapley Additive exPlanations
5	LIME	Local Interpretable Model-Agnostic Explanations
6	BMI	Body Mass Index
7	EHR	Electronic Health Records
8	AUC	Area Under Curve
9	ROC	Receiver Operating Characteristic
10	UML	Unified Modeling Language
11	ER Diagram	Entity Relationship Diagram

CHAPTER-I

INTRODUCTION

1.1About The Project

The evolution of the technology associated with Artificial Intelligence (AI) in the healthcare sector has led to changes in the methods through which diseases can be detected, diagnosed, and treated. The AI-based healthcare system can identify diseases in their early stages, provide personalized treatment plans for individuals and enable clinical decisions based on the effective analysis of huge volumes of patient data. The traditional disease prediction model takes into account only one disease in the healthcare sector with its limited clinical dimensions. Therefore, this paper presents the Smart Explainable AI Model for Predicting and Assessing Risks of Multiple Diseases at an Early Stage.

The proposed model aims to predict multiple diseases using one single framework. For this purpose, the system will make use of structured Electronic Health Records (EHR) and some benchmark medical datasets like Pima Indians Diabetes Dataset, Cleveland Heart Disease Dataset, and Wisconsin Breast Cancer Dataset. The proposed framework adopts a stacking ensemble algorithm, which will incorporate Random Forest, XGBoost, and Logistic Regression algorithms as base learners and Logistic Regression as meta-learner. Furthermore, the suggested model incorporates approaches of Explainable AI (XAI) such as SHAP and LIME that assist in obtaining greater transparency and interpretability. As such, with the aid of precise predictions and interpretability, it becomes possible to engage in proactive healthcare and make informed decisions related to diabetes, heart disease, and breast cancer.

CHAPTER-II
SYSTEM ANALYSIS

2.1 Existing System

Currently used disease prediction models in healthcare only focus on predicting one disease based on traditional machine learning techniques like SVM, Decision Tree, Naïve Bayes, and Logistic Regression. These models only consider structured clinical variables like blood pressure, glucose level, cholesterol levels, and age in their predictive models. While these models perform well in predicting diseases, they have various shortcomings that impact their usage in the field of healthcare.

Firstly, most of the currently used predictive models lack capability for predicting multiple diseases in a single model. For instance, separate models are created to predict diabetes, heart disease, or cancer. In addition to adding complexity to computations, such models make them hard to use in practice. Secondly, most of these models work as black box systems, making them difficult to interpret by healthcare practitioners.

Current predictive models do not provide probability estimates, severity, personalized health care advice for the predicted disease. These models also ignore other significant features that include demographic, behavior-related, and life-style factors.

Disadvantages

- Can only predict one disease at once.
- Dependent on structured clinical information.
- Inferior accuracy and generalization ability.
- Lacks transparency and explainability.
- Cannot offer tailored suggestions or risk assessment.
- Difficult to implement in actual medical settings.

2.2 Proposed System:

Proposed Smart Explainable AI Framework – “Smart Explainable AI Model for Personalized Disease Risk Assessment and Early Disease Prediction” – is expected to address issues in current healthcare prediction systems. The prediction model framework is meant to provide predictions of multiple diseases like diabetes, heart diseases, and breast cancer using one intelligent AI framework.

The prediction model proposed uses EHR with other health factors including clinical, demographic, behavioral, and life-style data. It employs a hybrid stacking ensemble learning framework with Random Forest, XGBoost, and Logistic Regression models as base learners, while Logistic Regression serves as meta-learner in the prediction framework. This ensures better accuracy, reliability, and robustness for multi-disease prediction.

Explainable AI frameworks such as SHAP and LIME are integrated into this prediction model framework for better understanding of predictions made. While SHAP provides global importance ranking of variables, LIME enables interpretation of individual predictions using local explainability. Variables like glucose, BMI, age, concave points_worst, and area_worst are some of the factors that contribute to disease predictions made using this model.

Advantages

- Can predict several diseases using one system.
- Has hybrid ensemble learning to improve precision.
- Has Explainable Artificial Intelligence to enhance transparency.
- Personalizes medical advice.
- Assesses disease risks and severity.
- Boosts trust and usability in medical systems.

2.3. Feasibility Study

Requirements

Hardware Requirements

- Operating System : Windows 7 or7+
- processor : 13/Inter Processor
- RAM : 4GB (min)
- Hard Disk : 500GB
- Key Board : Standard Windows Keyboard
- Mouse : Python 3.6+
- Monitor : SVGA

Software Requirements

- Operating System : Windows 10
- Server side Script : 8GB (min)
- IDE : 160GB

2.4. Feasibility Study

A feasibility study analyses the system under consideration for its practicality, usefulness, and implementation possibility considering the available resources and limitations.

1. **Technical Feasibility:** It is technologically possible to implement the system since it is based on popular machine learning libraries and common healthcare data sets. Technologies such as Python, Scikit-learn, XGBoost, SHAP, and LIME allow implementing the framework efficiently.

2. **Economic Feasibility:** The implementation of the proposed system does not involve the use of costly resources since it relies only on open-source frameworks and freely accessible data sets. No specific expensive equipment or software is needed to implement the system.

3. **Operational Feasibility:** The system is simple and easy to use since it was developed to facilitate decision-making for medical experts. Its use together with Explainable AI technology makes the system easier to use since it allows understanding predictions better.

4. **Schedule Feasibility:** There is an opportunity to complete the project on time because all the necessary data sets, frameworks, and technologies required by the project are available.

CHAPTER – III
SYSTEM DESIGN

3.1. Module Description:

The suggested “Smart Explainable AI Model for Personalized Disease Risk Assessment and Early Disease Prediction” system uses a modular approach to facilitate scalability, flexibility, interpretability, and efficiency in disease prediction. This system will utilize machine learning, deep learning, ensemble learning, and XAI techniques to predict various diseases by analyzing their medical, lifestyle, and demographic information.

3.1 MODULE DESCRIPTION

1. User Interface Module

Description

It is an interface that facilitates communication between the user and AI system. This module enables patients, doctors, or healthcare experts to provide patient medical information and analyze the results of the prediction.

Inputs

- Patient age
- Patient gender
- Blood pressure
- Glucose level
- Cholesterol
- Lifestyle factors
- Health history
- Outputs
- Prediction results

- Risk assessment report
- Personalized suggestions

Functions

- User registration and login
- Enter patient medical information
- Health record uploading
- Presentation of predictive analysis report
- Disease prediction results
- Explainable AI presentation

2. Data Collection Module

Description

This module is responsible for collecting and storing data related to healthcare.

Functions

- Medical data collection and storage
- Demographic and lifestyle data integration
- Collection of multi-disease dataset
- Securely store patient data

Dataset Used

- Diabetes Dataset
- Heart Disease Dataset
    - Cancer Dataset

Benefits

- Consolidated data processing
- Enhanced data consistency
- Effective data storage

3. Data Preprocessing Module

Description

Medical datasets frequently suffer from missing values, noise, and errors. This module prepares the raw data prior to training the machine learning algorithms.

Functions

- Handling Missing Values
- Removal of Duplicates
- Normalization of data
- Feature encoding
- Feature scaling
- Outliers detection

Techniques Utilized

- Min-max Scaling
- Standardization (StandardScaler)
- Label Encoding
- Balancing Data (SMOTE)

Output

The formatted data set is prepared for machine learning models.

4. Feature Engineering Module Description

In this module, relevant attributes are extracted from healthcare data for accurate predictions.

Functions

- Selection of features
- Analyzing correlation between variables
- Dimensionality Reduction
- Identifying important medical factors

Techniques

- Heatmap for correlations
- Recursive feature elimination (RFE)
- PCA (optional)

Advantages

Makes the model simple Enhances predictive performance Eliminates redundant features.

5. Machine Learning Prediction Module

Description

In this module, various machine learning algorithms are trained for predicting diseases.

Algorithms

- Random forest
- XGBoost
- Decision tree
- Logistic Regression
- Support Vector Machine (SVM)

Functions

- Training the ML models
- Predicting disease
- Comparing model performances
- Producing probability scores

Output

Predicted disease category.

6. Deep Learning Module

Description

The module uses Artificial Neural Networks to model complicated relations between features in health care data.

Methods Used

- Artificial Neural Network (ANN)
- Deep Neural Network (DNN)

Functions

- Detection of hidden connections between diseases
- Accuracy improvement
- Consideration of complicated nonlinear interactions

Advantages

- Improved learning capability
- Higher prediction performance
- Appropriate for processing large amounts of data.

7. Ensemble Learning Module

Description

The module is an intelligent one that uses the results of multiple machine learning and deep learning models to improve its performance.

Functions

- Combining predictions made by different algorithms
- Prevention of overfitting
- Increase robustness and stability Improving accuracy and generalization capability

Types of Ensemble Learning

- Voting
- Stacking
- Weighted averaging

Advantages

- Improvement in prediction accuracy
- Improved performance
- Disease classification.

8. Explainable Artificial Intelligence (XAI) Module

Description

The module provides an explanation of how the system reached the conclusion about the particular disease. Thus, improving transparency and reliability of AI-driven health care solutions.

Methods Used

- SHAP (SHapley Additive Explanations)
- LIME (Local Interpretable Model-Agnostic Explanations)

Functions

- Explaining feature importance
- Explainability of decision process
- Creation of reports

Example

The system will provide the following explanations:

- Glucose level high contributes to 40%.
- Body mass index high contributes to 20%.
- Smoking contributes to 15%.

9. Personalized Recommendation Module

Description

The system gives personalized recommendations based on the results of predictions**.**

Recommendation Types

- Diet plan suggestions
- Workout routines
- Health lifestyle suggestions
- Doctor visit recommendations

Advantages

- Support for preventive health care
- Early detection
- Increased patient awareness

3.2 Entity Relationship Diagram

Description

Entity Relationship (ER) diagram shows the relationship among users, patients' information, disease prediction results, AI algorithms/models, and recommendation system.

Entities Involved

1. User

Attributes:

- User ID (Primary Key)
- Name
- Email
- ID
- Password
- Role

2. Patient_Data

Attributes:

- Patient_ID (Primary Key)
- Age
- Gender
- Blood Pressure
- Glucose level
- Cholesterol
- BMI
- Lifestyle habits

3. Disease_Prediction

Attributes:

- Prediction_ID (Primary Key)

Disease Name

- Risk Score
- Prediction Result
- Date of prediction

4. AI Model

Attributes:

- Model_ID
- Model_Name
- Accuracy
- Type of Algorithm

5. Explainability Report

Attributes:

- Report_ID
- SHAP Score
- LIME_Explanation
- Important Features

6. Recommendation

Attributes:

- Recommendation_ID
- Diet Plan

- Exercise Plan
- Medical Advice

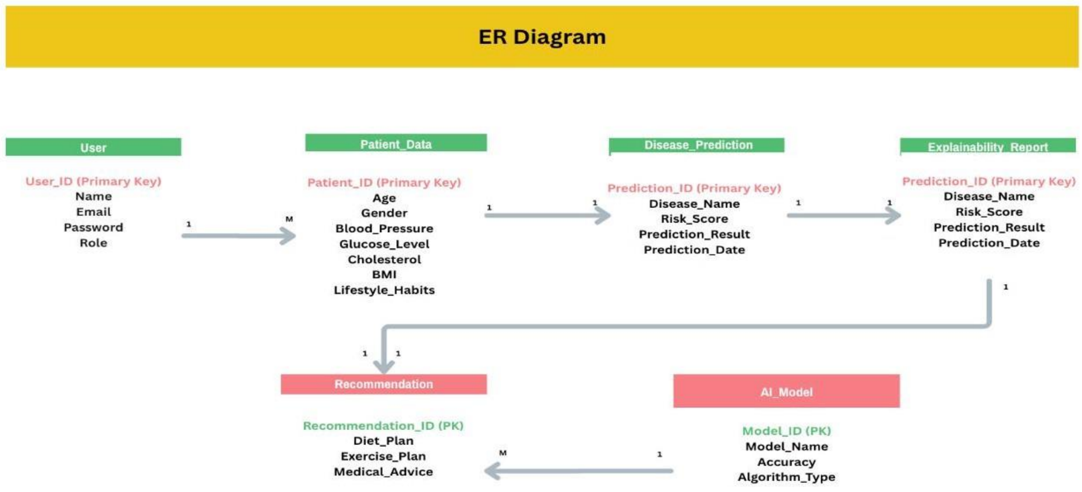

Figure 3.2.1. Entity Relationship Diagram

The ER diagram shows the schema design of the Smart Explainable AI Model for Disease Risk Assessment & Early Prediction. User is an entity that includes data such as User_ID, Name, Email, Password, and Role. There exists a one to many relation between entities User and Patient Data; that is, Patient Data is an entity that includes important information about the user, including Age, Gender, Blood Pressure, Glucose, Cholesterol, Body Mass Index, and Lifestyle. This entity uses all this data from Patient_Data and generates disease prediction results, risk score, and prediction date in Disease_Prediction entity. Explainability_Report stores outputs from Explainable AI that can be used by users to understand disease prediction results. Recommendation generates personalized dietary plan, exercise, and other medical recommendations based on predictions. The Machine Learning Model Details of each model used in the system is stored in AI_Model.

3.3 UML DIAGRAMS

The Unified Modeling Language (UML) diagrams are utilized to depict the structure and operation of the suggested artificial intelligence-based health care system. Such diagrams ensure better understanding of the interaction between different elements of the system, making sure that there is clarity while designing and developing the same. In the current study, use of UML diagrams will be essential in depicting various aspects of the architecture of the system related to disease prediction, explainability, and recommendations.

3.3.1 Use Case Diagram

Actors

- Patient
- Doctor
- Admin

Use Cases

- Login/Registration
- Provide Medical Information
- Disease Prediction
- Explain Report
- Recommendation Generation
- Download Report
- Maintain Dataset
- Training of AI Model

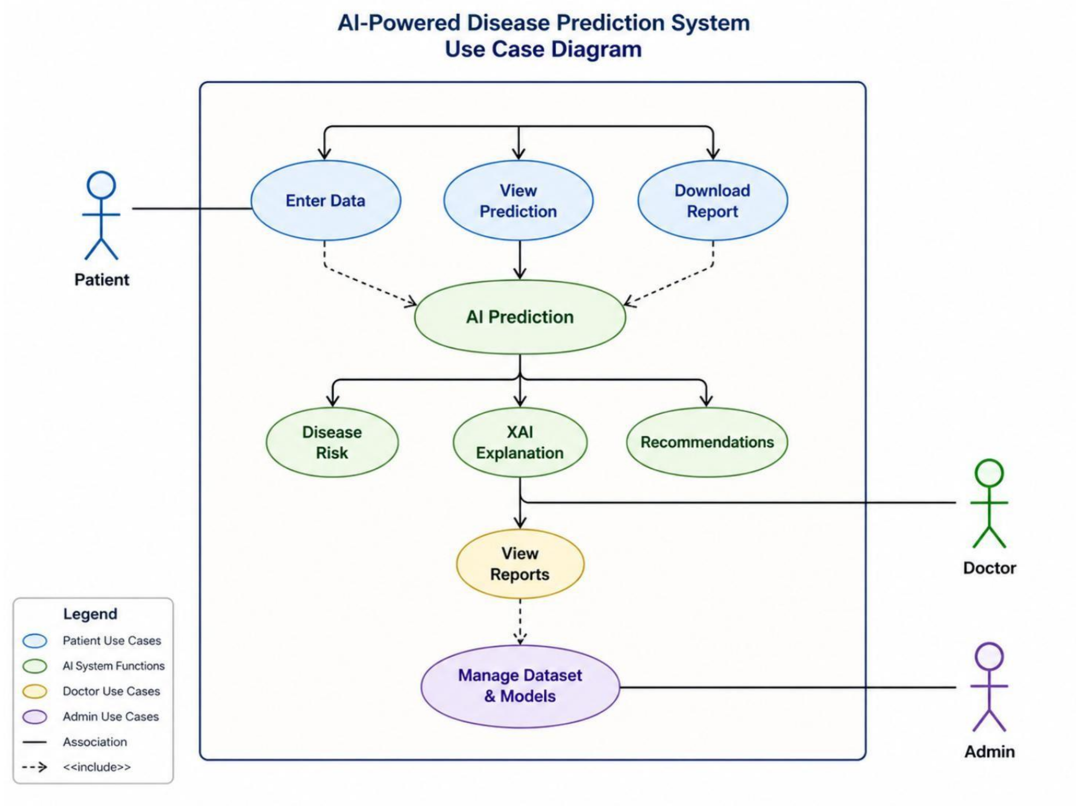

Figure 3.3.1.1 Use Case Diagram

The use case diagram above depicts an AI disease predictor with interactions taking place among three primary actors: patient, doctor, and admin. It begins with a patient inputting their health information into the system and making use of predictions and downloadable report outputs. These are some of the interactions that correspond to the main use case, which is "AI Prediction," where information inputted into the AI system is used to give disease predictions, explanations through XAI, and personalized advice on diseases. The XAI explanation plays the role of explaining the AI's prediction. In the system, there is a secondary use case whereby doctors view reports generated using the predictions from the previous use case. On the other hand, admin manages datasets and models within the system.

3.3.2 Class Diagram

Classes Used

1. **User**

   The User class handles user logins and logouts.

   Attributes:
   - userID
   - name
   - email

   Methods:
   - login()
   - logout()

2. **Patient**

   The Patient class contains the data of the patient and displays the result of the predictions. It inherits the User class.

   Attributes:
   - medicalData

   Methods:
   - enterData()
   - viewPrediction()

3. **AIModel**

   The AIModel class uses machine learning to predict diseases.

   Attributes:
   - modelName
   - accuracy

   Methods:
   - trainModel()
   - predictDisease()

4. **Explainability**

   The Explainability class provides explainable artificial intelligence results using the SHAP and LIME approaches.

   Attributes:
   - shapScore
   - limeExplanation

Methods:
- ➢ generateSHAP()
- ➢ generateLIME()

5. **Recommendation**

The Recommendation class provides advice on maintaining a healthy lifestyle.

Attributes:
- ➢ advice

Methods:
- ➢ generateAdvice()

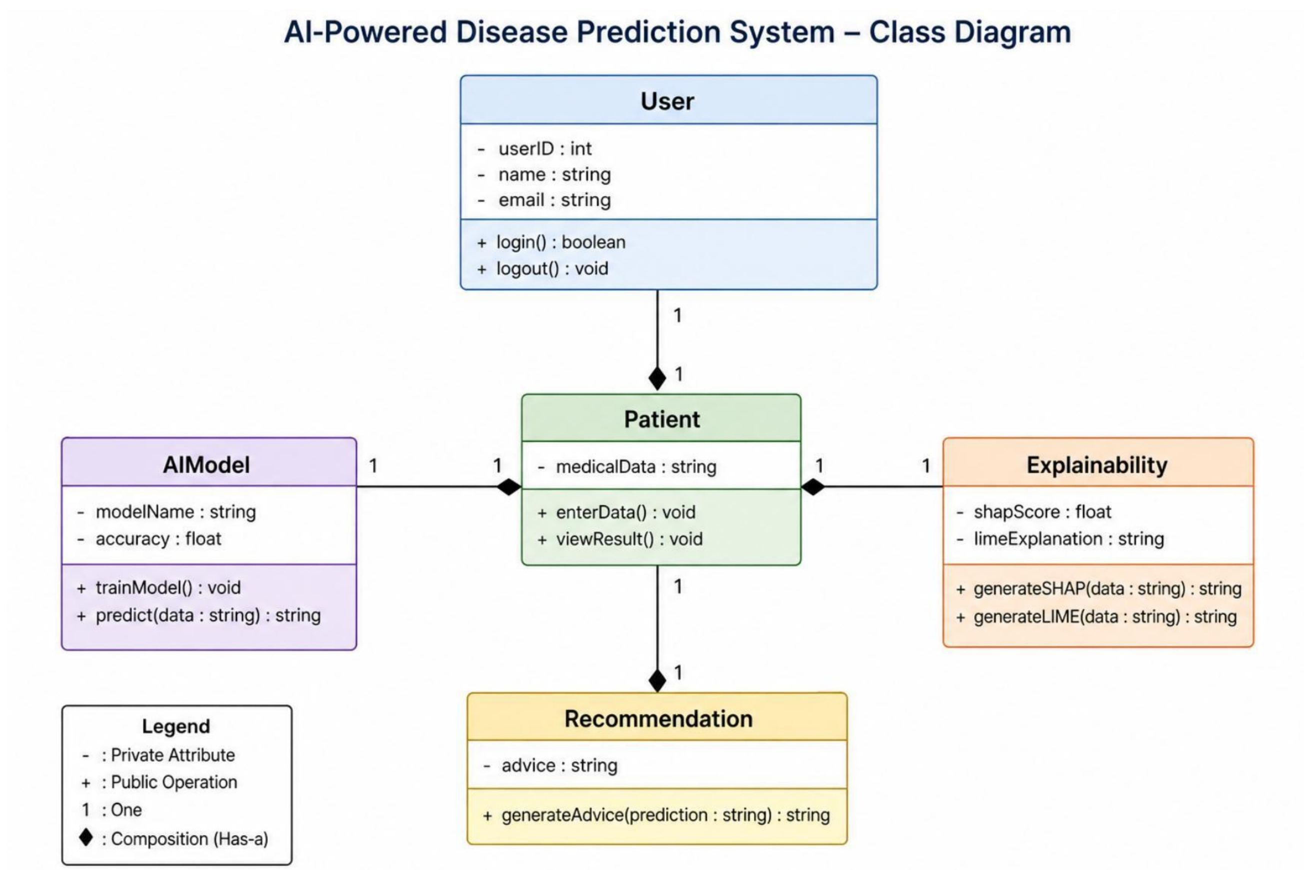

Figure 3.3.2.1 Class Diagram

3.3.3 Activity Diagram

The Activity Diagram illustrates the flow of activities in the suggested system. The diagram explains the steps of activities that occur while forecasting diseases and creating customized suggestions with Explainable Artificial Intelligence.

Explanation of Activity Flow

1. **Start :** The activity flow starts with the application opening on the user side.

2. **User Login :** The user logs into the application through valid identification to use the prediction system.

3. **Medical Data Input :** Medical data such as age, blood pressure, glucose level, BMI, and lifestyle choices are entered by the patient.

4. **Data Preprocessing :** Data preprocessing is done by managing missing values, filtering noise, and formatting the data to be used in machine learning algorithms.

5. **Feature Selection :** The system selects medical data features from the data set.

6. **Predicting Diseases :** Using the machine learning model, the diseases like diabetes, heart disease, or kidney disease can be predicted from patient information.

7. **Generating XAI Explanations** : The Explainable Artificial Intelligence module makes use of SHAP and LIME techniques to give an explanation for the prediction.

8. **Generating Recommendations:** Based on the prediction output, recommendations are made for health issues and precautions to be followed by the patient.

9. **Presenting Results** : Finally, results along with the explanations and recommendations are shown to the user.

10. **End** : The end stage is achieved after displaying all outputs to the user.

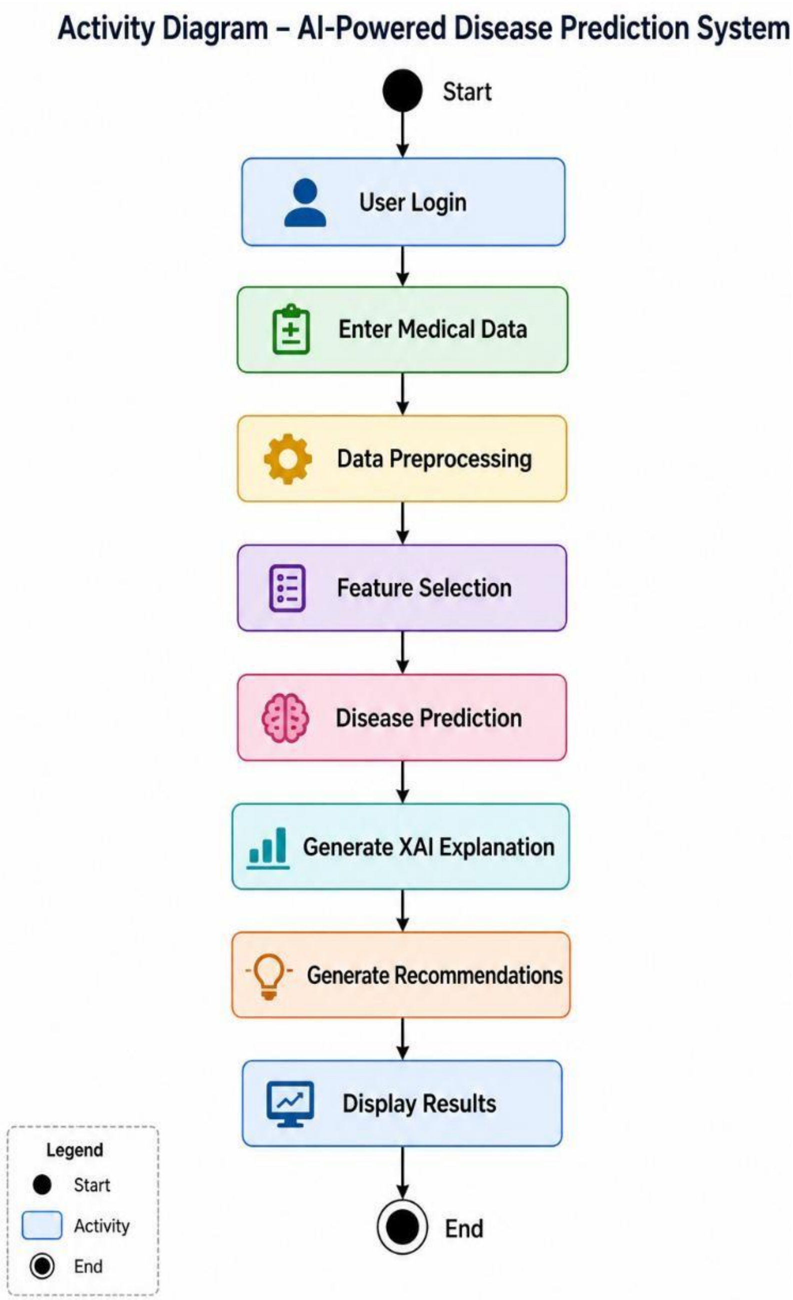

Figure 3.3.3.1 Activity Diagram

3.3.4 Sequence Diagram

The Sequence Diagram illustrates the interaction between the patient and various system modules while making predictions about diseases. It provides an overview of how data is processed from input through prediction to explanation generation and recommendation presentation.

Sequence Flow Description

1. **Login of Patient :** Firstly, the patient logs in to the system with appropriate login credentials.

2. **Input of Medical Details**: Once logged in, the patient inputs their medical and lifestyle data such as age, blood pressure, glucose levels, BMI, among others.

3. **Preprocessing of Inputted Data:** The system directs the patient's entered data to the Preprocessing Module, where missing data is imputed, unnecessary data is eliminated, and the data is formatted appropriately.

4. **Transmission of Preprocessed Data to AI Model**: The Preprocessing Module transfers the preprocessed data to the AI Model for processing.

5. **Disease Prediction**: Using the information entered by the patient, the algorithm predicts the possible disease the patient may suffer from.

6. **Explainable AI Report Creation:** The prediction results will be then sent to the XAI component which creates SHAP and LIME reports about prediction results.

7. **Generate Recommendations:** Based on the gained knowledge from predictions and explanations, recommendations for patients are created by the Recommendations component.

8. **Displaying Results:** The last step is the displaying of disease prediction results, explainability reports, and recommendations to the user.

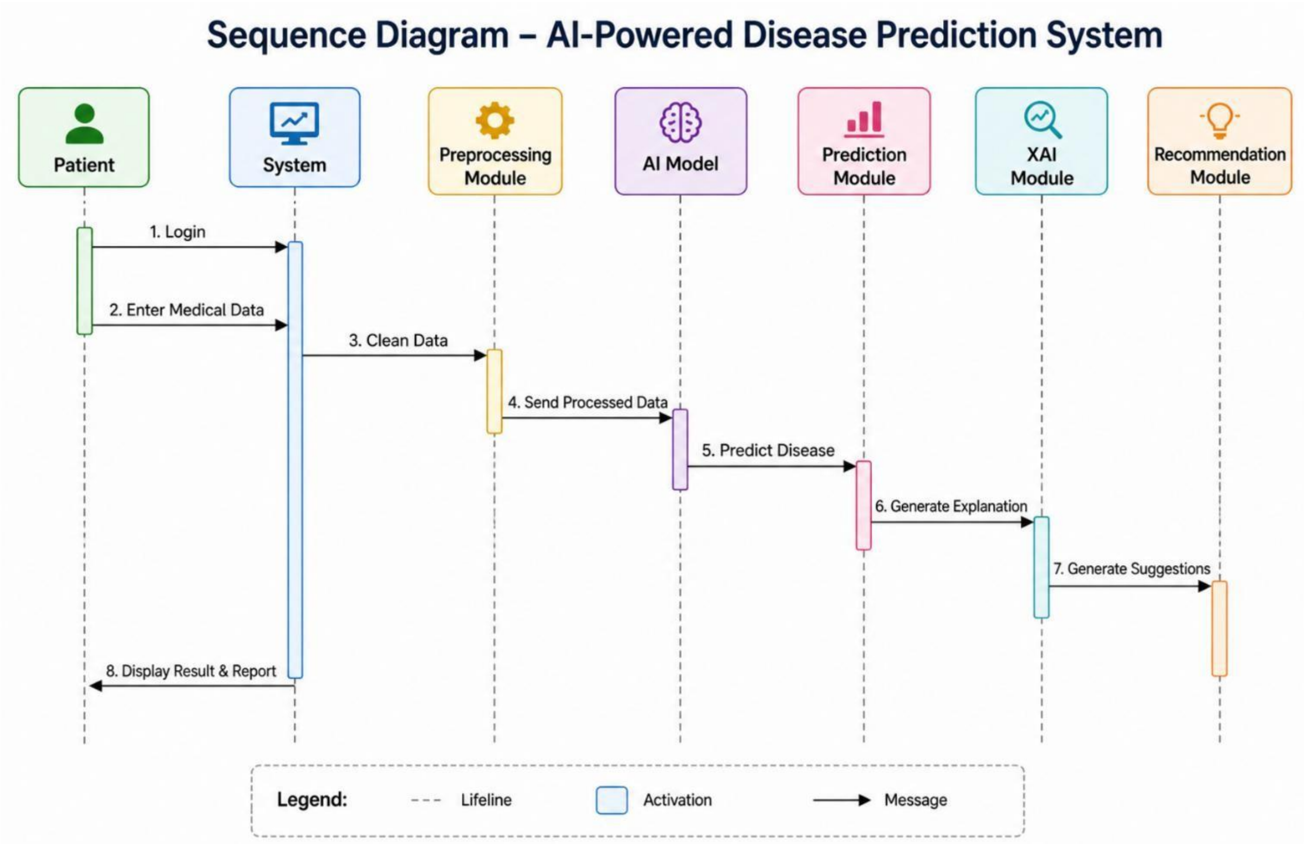

Figure 3.3.4.1 Sequence Diagram

The figure above demonstrates the detailed workflow of an AI-based disease prediction system through various components. The procedure starts when a patient signs in to the system and input their medical data. This information is forwarded to the preprocessing module, which cleans the input data. Further, this processed data is forwarded to the AI module to conduct the disease prediction process. Following this, the prediction output is delivered to the prediction module, responsible for coordinating the process further. After the prediction process, the explanation of this output is generated by the XAI (Explainable AI) module. Further, the recommendation module offers recommendations on predicted disease conditions. Lastly, the output, containing prediction results, explanations, and recommendations, are provided to the user in form of a report by the system.

CHAPTER – IV
SYSTEM IMPLEMENTATION

4.1 Choice of Language

The implementation of the project “Smart Explainable AI Model for Personalized Disease Risk Assessment and Early Disease Prediction” needs effective programming languages, machine learning libraries, visualization methods, and web development frameworks to accurately predict diseases, make them explainable, and interact with users.

4.1.1 Choice of Programming Language – Python

Python is chosen as the main language in this project due to its easy learning curve, versatility, and extensive support for Artificial Intelligence and Machine Learning.

Reasons Behind Choosing Python Language

- Simpler syntax compared to other languages
- Strong support for Machine Learning and Deep Learning applications
- Support for advanced libraries for data analysis and visualization
- Large developer community and documentation support
- Efficient and fast development and testing process

Applications

- Data pre-processing
- Development of disease prediction models
- Implementation of ensemble learning
- Application of Explainable AI models (SHAP and LIME)
- Visualization creation
- Integration with backend

4.1.2 Libraries Used

1. Pandas
Used for:
Cleaning of data Dealing with missing values Manipulation of data Processing of CSV dataset
Code Snippet:
import pandas as pd data = pd.read_csv("dataset.csv")

2. NumPy
Used for:
Performing mathematical calculations Matrix operations Numerical computations Code Snippet:
import numpy as np arr = np.array([1,2,3])

3. scikit-learn
Application:
- Machine learning models
- Learning algorithms
- Model evaluation
- Ensemble modeling

Algorithms Utilized:
- Random Forest Classifier
- XGBoost
- Logistic Regression
- Support Vector Machines

Code Snippet
from sklearn.ensemble import RandomForestClassifier

4. XGBoost
Application:
High-speed gradient boosting algorithms Disease prediction accuracy Importance of features
Code Snippet:
from xgboost import XGBClassifier

5. SHAP (SHapley Additive Explanation)
Application:
Explainable Artificial Intelligence Features contribution to model Interpretation of model

Objective:
Allows users to comprehend the reasons behind a predicted disease result.
Code Snippet:
```
import shap shap_explainer = shap.TreeExplainer(model)
```

6. LIME (Local Interpretable Model-Agnostic Explanation)
Application:
Local explanations of predictions Specific patient-level interpretations

Code Snippet:
```
from lime.lime_tabular import LimeTabularExplainer
```

4.1.3 Frontend Technologies
HTML
- Web Page Design
- Patient Data Form Creation

CSS
- Webpage Styling
- Responsive UI Development

JavaScript
- Interactive Webpages
- Form Validation
- Intuitive UI Interface

4.1.4 Backend Framework Used
Flask
Flask is used as the backend framework for the use of machine learning algorithms.

Advantages
- Lightweight Framework
- Ease Of Integration With Python-Based ML Algorithms
- Quick Processing Speed
- Web Deployment Support

Example:
```
from flask import Flask
app = Flask(__name__)
```

4.2 Screen Shots

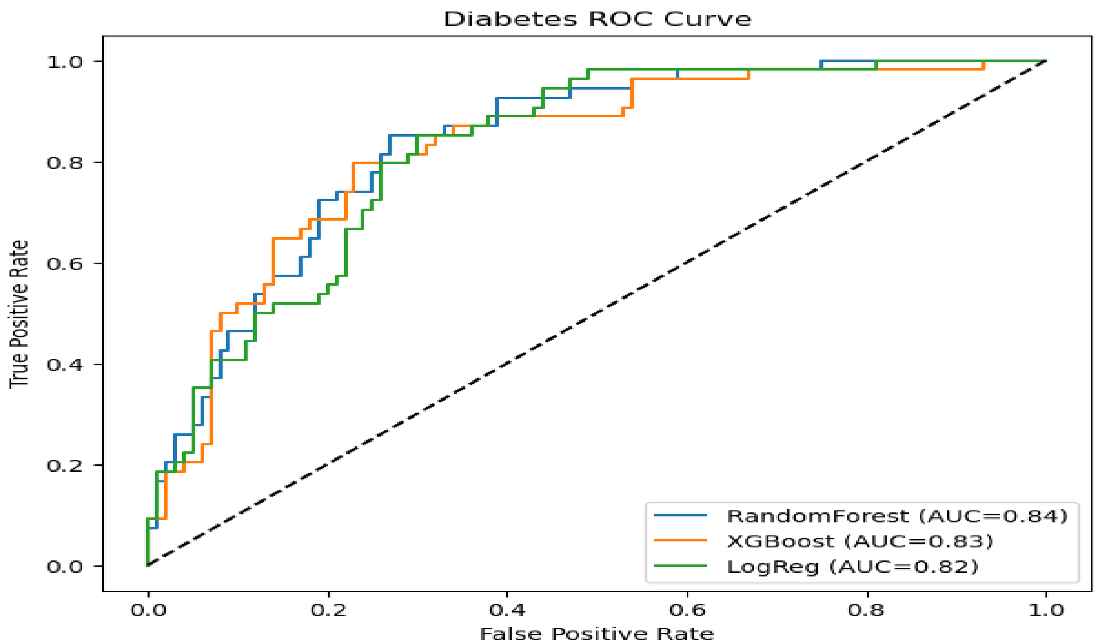

Figure 4.2.1. Diabetes ROC Curve

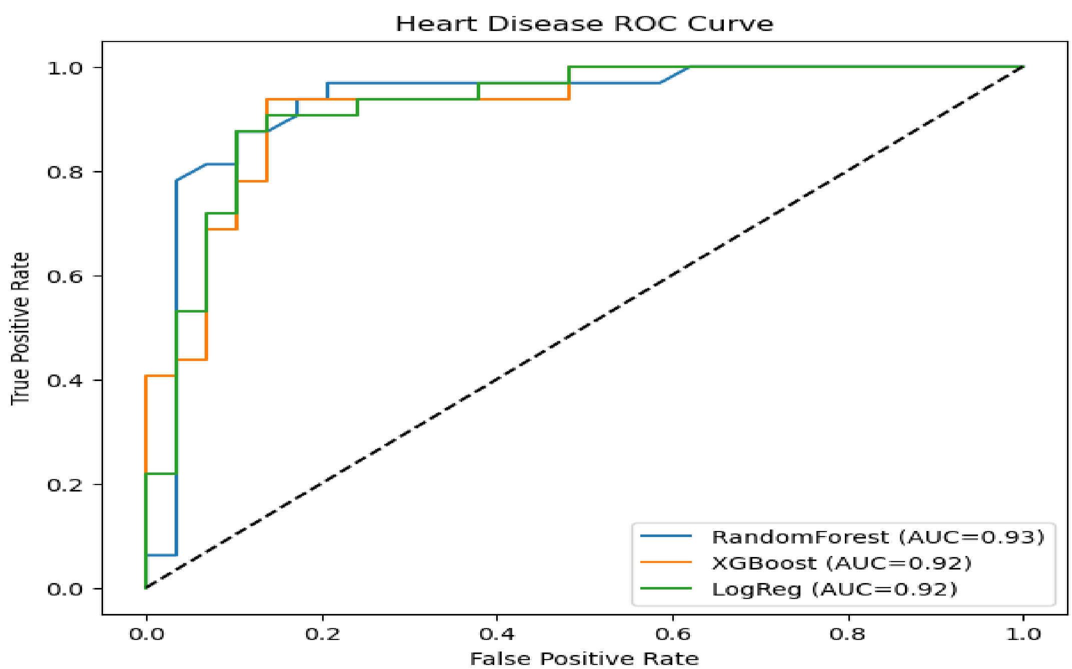

Figure 4.2.2. Heart Disease ROC Curve

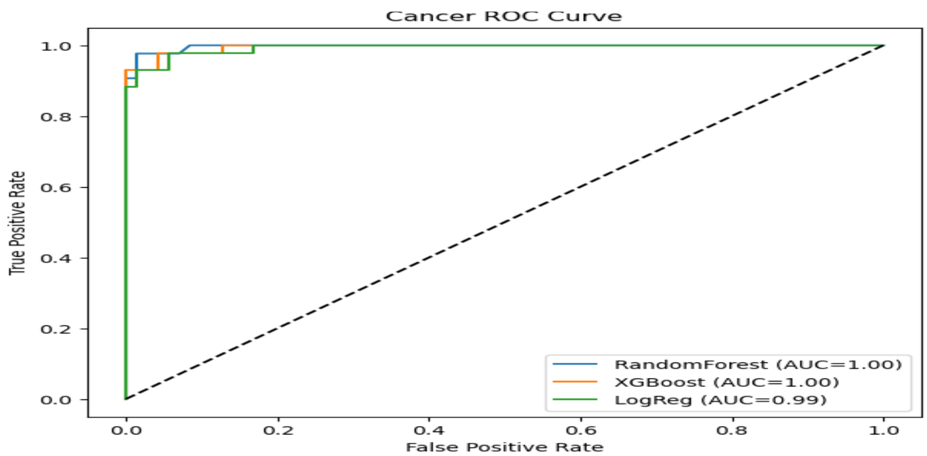

Figure 4.2.3. Cancer ROC Curve

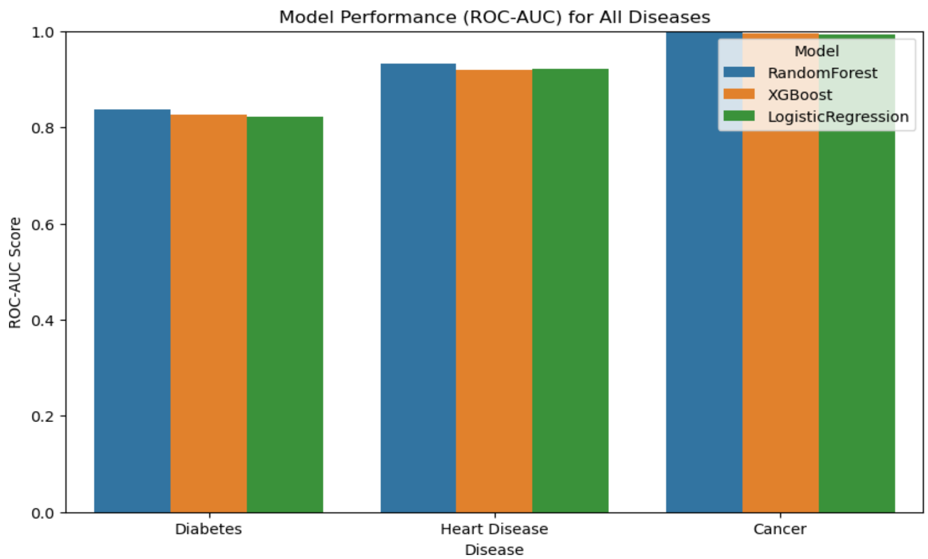

Figure 4.2.4. Model performance for all diseases

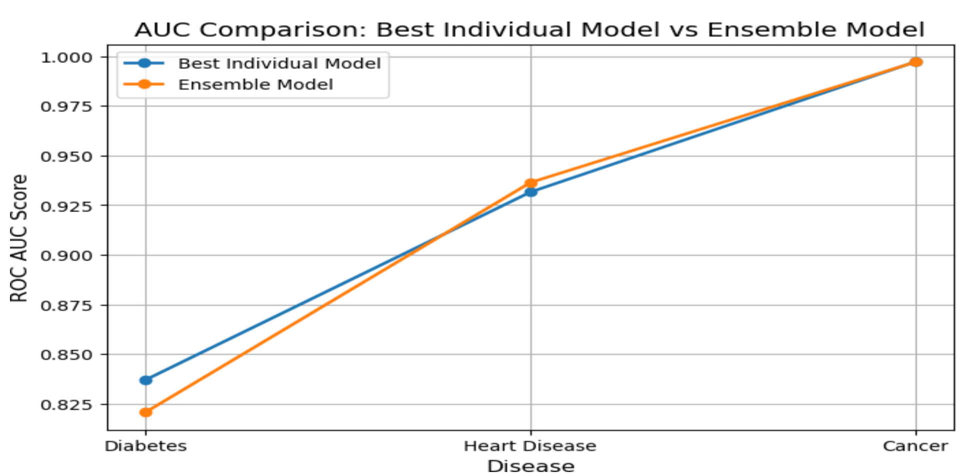

Figure 4.2.5. AUC comparison of Best Individual Model vs Ensemble Model

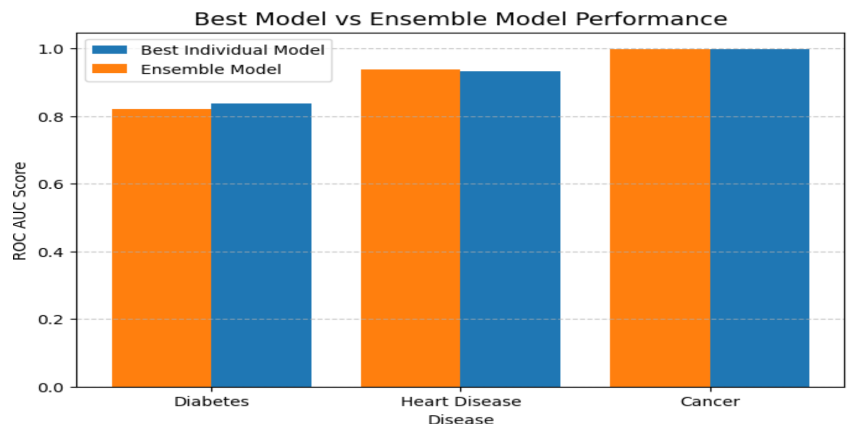

Figure 4.2.6. Best Model vs Ensemble Model Performance

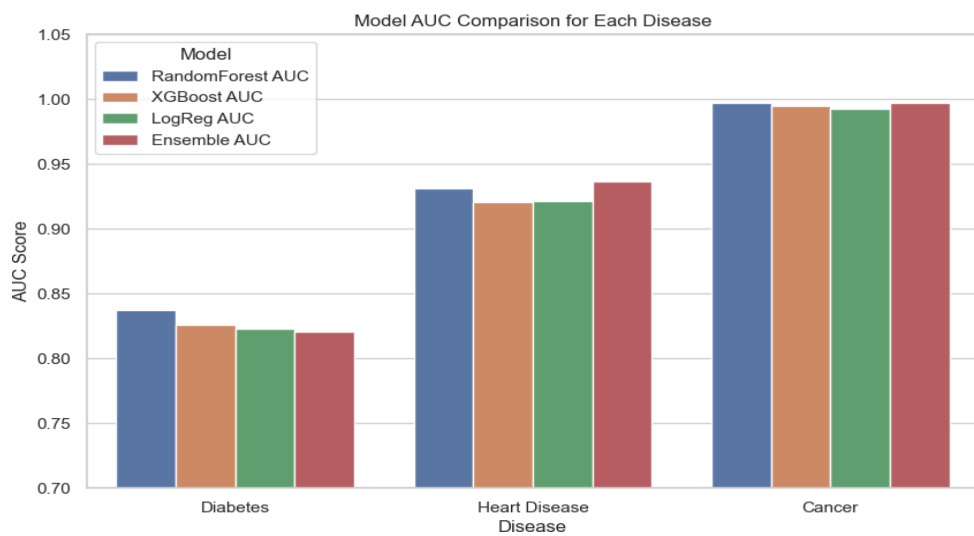

Figure 4.2.7. Model AUC comparison for Each Disease

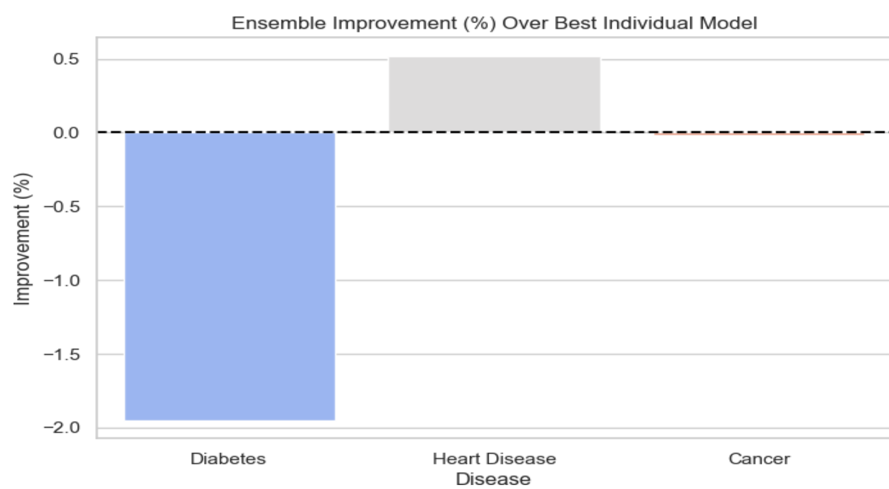

Figure 4.2.8. Ensemble Improvement (%) Over Best Individual Model

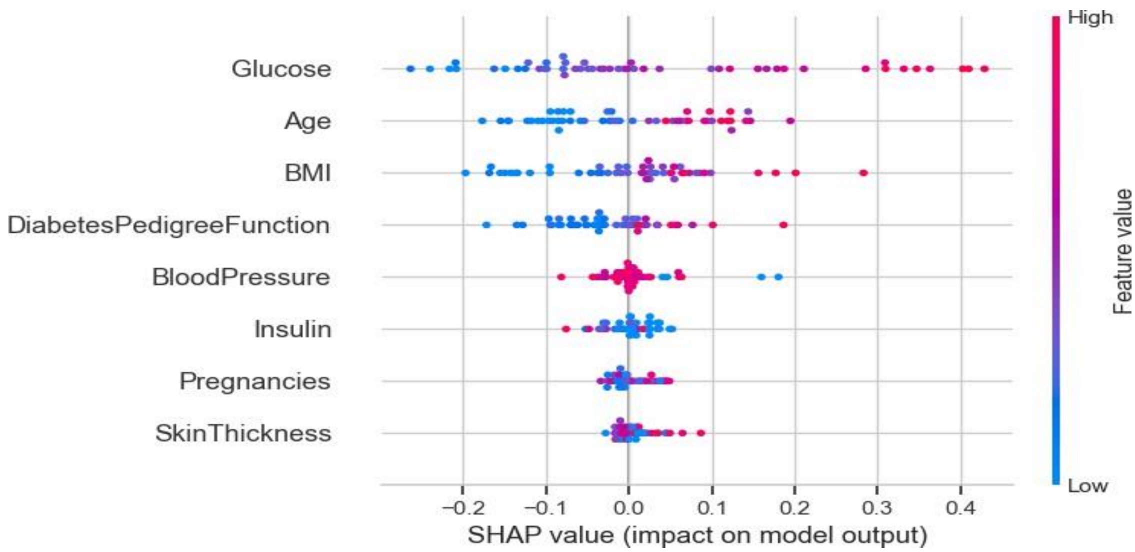

Figure 4.2.9. SHAP Summary Plot for Feature Importance in Predicting Diabetes

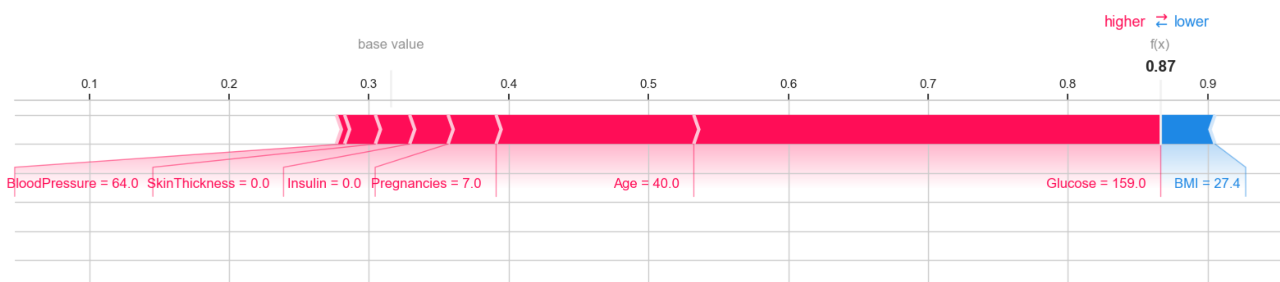

Figure 4.2.10. SHAP Force Plot for Individual Prediction in Predicting Diabetes

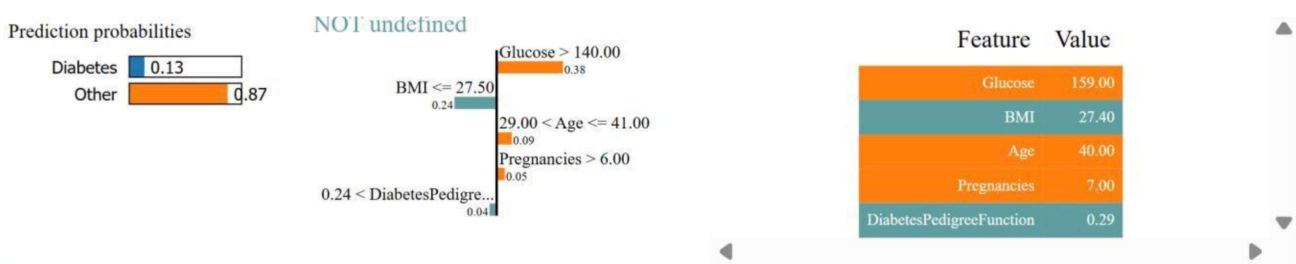

Figure 4.2.11. Visualization of LIME Explanation for Predicting Personalized Diabetes Risk

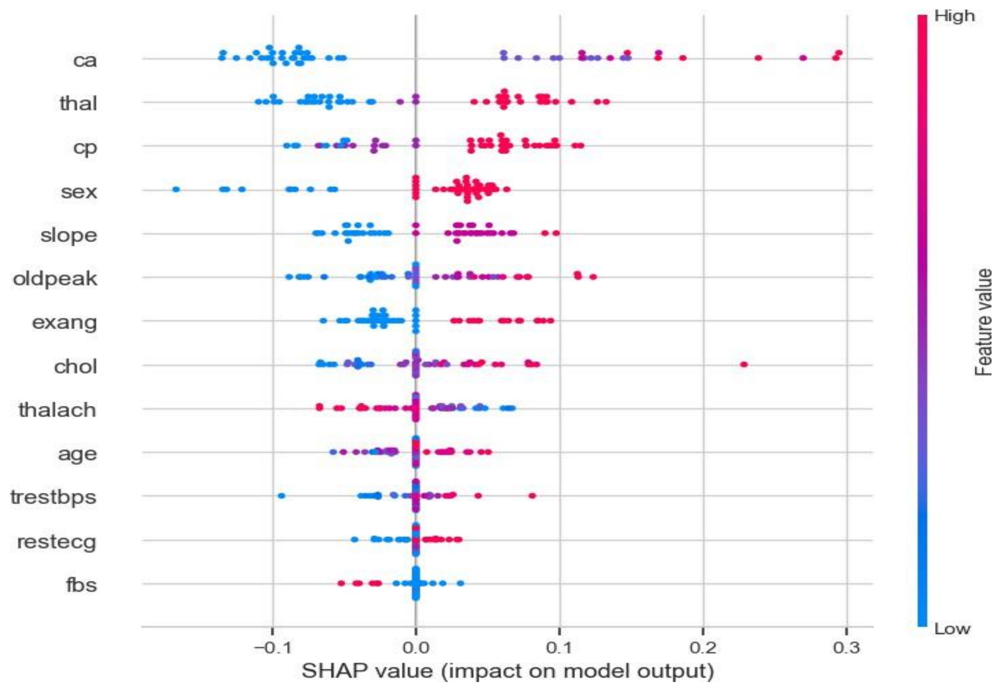

Figure 4.2.12. SHAP Summary Plot for Feature Importance in Predicting Heart Disease

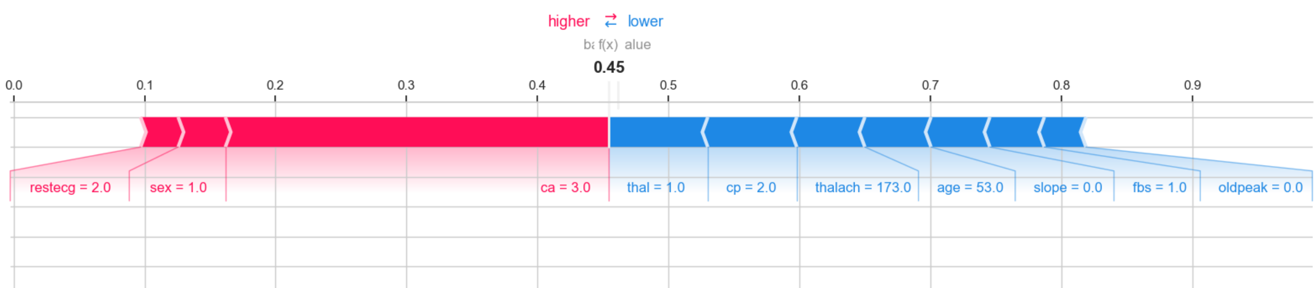

Figure 4.2.13. SHAP Force Plot for Individual Prediction in Predicting Heart Disease

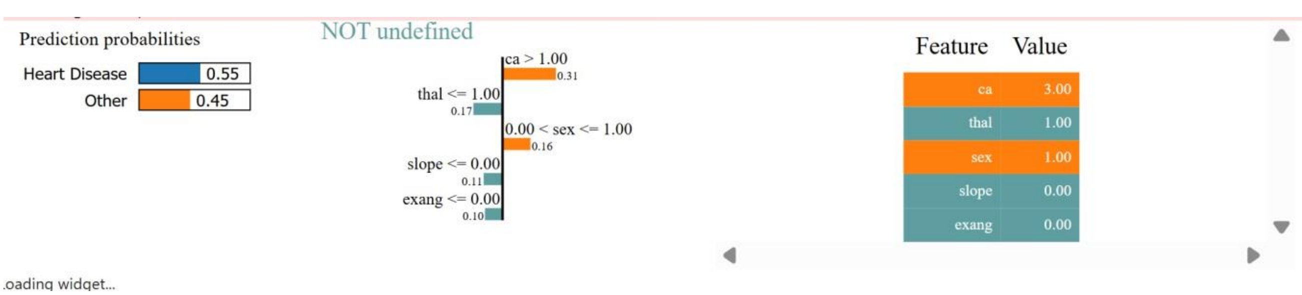

Figure 4.2.14. Visualization of LIME Explanation for Predicting Personalized Heart Disease Risk

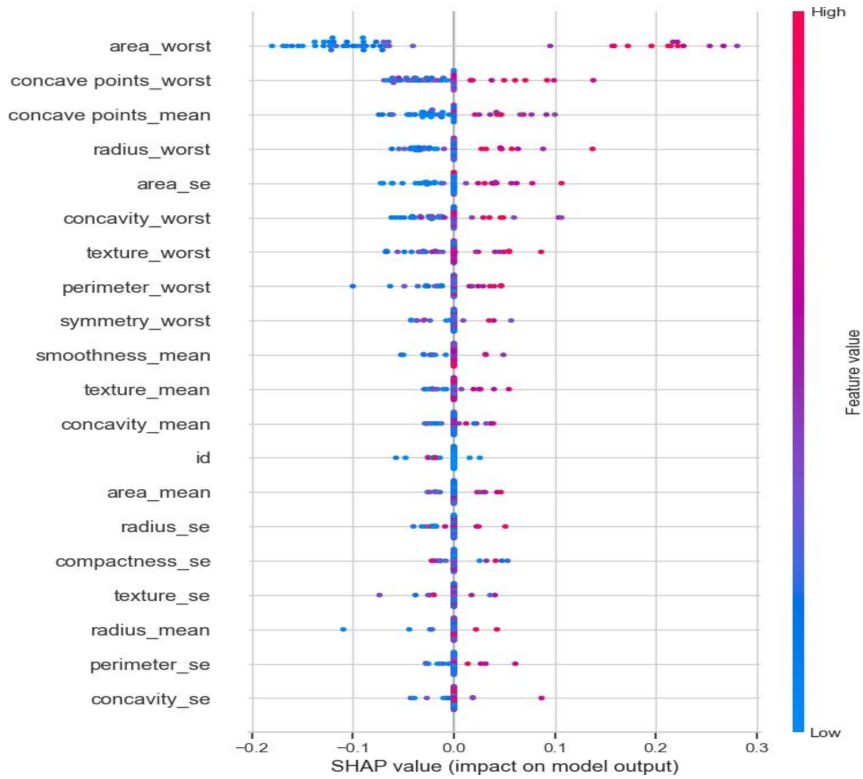

Figure 4.2.15. SHAP Summary Plot for Feature Importance in Predicting Cancer

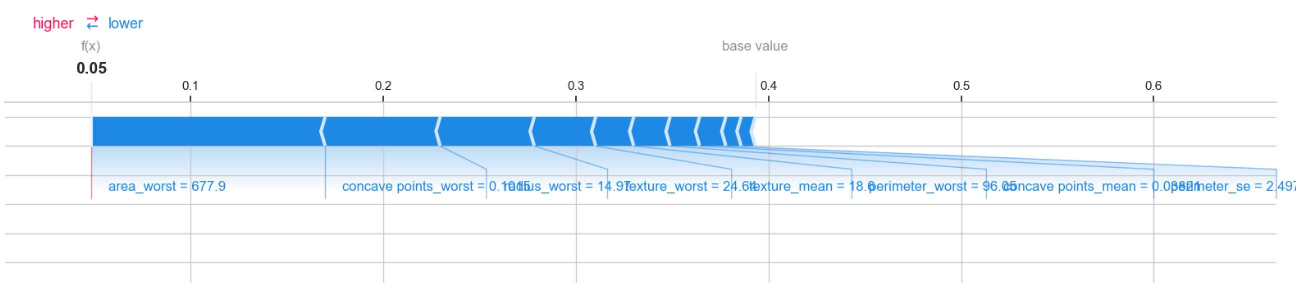

Figure 4.2.16. SHAP Force Plot for Individual Prediction in Predicting Cancer

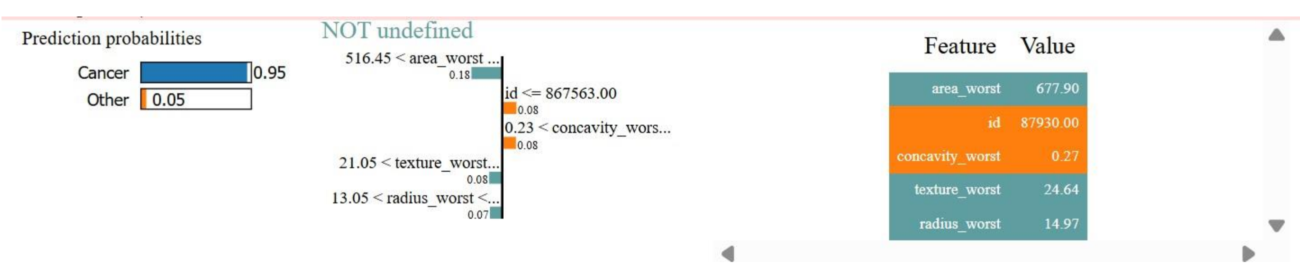

Figure 4.217. Visualization of LIME Explanation for Predicting Personalized Cancer Risk

	Patient	Disease	Risk Score	Key Features	Recommendation
0	0	Diabetes	0.87 (High)	Glucose, Age, Pregnancies	Reduce carbs, exercise, medical consultation
1	1	Diabetes	0.13 (Low)	Glucose, Age, DiabetesPedigreeFunction	Reduce carbs, exercise, medical consultation
2	2	Diabetes	0.09 (Low)	Age, Glucose, DiabetesPedigreeFunction	Reduce carbs, exercise, medical consultation
3	3	Diabetes	0.32 (Low)	BMI, Pregnancies, Age	Reduce carbs, exercise, medical consultation
4	4	Diabetes	0.06 (Low)	Glucose, BMI, BloodPressure	Reduce carbs, exercise, medical consultation
5	5	Diabetes	0.62 (Moderate)	DiabetesPedigreeFunction, Glucose, BMI	Reduce carbs, exercise, medical consultation
6	6	Diabetes	0.24 (Low)	Glucose, DiabetesPedigreeFunction, Age	Reduce carbs, exercise, medical consultation
7	7	Diabetes	0.74 (Moderate)	Glucose, BMI, Age	Reduce carbs, exercise, medical consultation
8	8	Diabetes	0.05 (Low)	Age, Glucose, BMI	Reduce carbs, exercise, medical consultation
9	9	Diabetes	0.86 (High)	Glucose, BMI, DiabetesPedigreeFunction	Reduce carbs, exercise, medical consultation

Figure 4.2.18. Results of Personalized Disease Risk Prediction and Recommendation Table

4.3 Sample Code

```
# Import libraries
import pandas as pd
import numpy as np

# Preprocessing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder

# SMOTE
from imblearn.over_sampling import SMOTE

# Models
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

# Evaluation
from sklearn.metrics import roc_auc_score, classification_report, confusion_matrix, accuracy_score,
roc_curve
import matplotlib.pyplot as plt

df_diabetes = pd.read_csv("Diabetes.csv")
df_diabetes.head()
df_heart = pd.read_csv("HeartDisease.csv")
df_heart.head()
df_cancer = pd.read_csv("BreastCancer.csv")
df_cancer.head()

from sklearn.model_selection import train_test_split
from imblearn.over_sampling import SMOTE
from sklearn.metrics import roc_auc_score, classification_report
from xgboost import XGBClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression

#  Detect target column name (Prevents KeyError)
possible_targets = ['Outcome', 'Diabetes', 'Class', 'target']
target_disease_col = None

for col in possible_targets:
    if col in df_diabetes.columns:
        target_disease_col = col
        break
print("Detected Diabetes Target Column:", target_disease_col)

#  Use detected target column
```

```
X_dia = df_diabetes.drop(columns=[target_disease_col])
y_dia = df_diabetes[target_disease_col]

#  Train-test split
X_train_dia, X_test_dia, y_train_dia, y_test_dia = train_test_split(
    X_dia, y_dia, test_size=0.2, random_state=42, stratify=y_dia)

#  Apply SMOTE
smote = SMOTE(random_state=42)
X_train_dia, y_train_dia = smote.fit_resample(X_train_dia, y_train_dia)
print("After SMOTE Diabetes:", X_train_dia.shape)

results = {}

#  RandomForest
rf = RandomForestClassifier(n_estimators=200, max_depth=8, class_weight='balanced',
random_state=42)
rf.fit(X_train_dia, y_train_dia)
rf_auc = roc_auc_score(y_test_dia, rf.predict_proba(X_test_dia)[:, 1])
results['Diabetes_RF_AUC'] = rf_auc

# XGBoost
xgb = XGBClassifier(n_estimators=200, learning_rate=0.05, max_depth=4, subsample=0.8,
random_state=42)
xgb.fit(X_train_dia, y_train_dia)
xgb_auc = roc_auc_score(y_test_dia, xgb.predict_proba(X_test_dia)[:, 1])
results['Diabetes_XGB_AUC'] = xgb_auc

#  Logistic Regression
lr = LogisticRegression(max_iter=500, class_weight='balanced')
lr.fit(X_train_dia, y_train_dia)
lr_auc = roc_auc_score(y_test_dia, lr.predict_proba(X_test_dia)[:, 1])
results['Diabetes_LR_AUC'] = lr_auc

print("\nDiabetes Model Performance:")
print(results)

X_heart = df_heart.drop('HeartDisease', axis=1)
y_heart = df_heart['HeartDisease']

X_train_h, X_test_h, y_train_h, y_test_h = train_test_split(X_heart, y_heart, test_size=0.2,
random_state=42)

# Train RandomForest
rf_h = RandomForestClassifier(random_state=42)
rf_h.fit(X_train_h, y_train_h)
rf_h_prob = rf_h.predict_proba(X_test_h)[:, 1]
```

```
rf_h_auc = roc_auc_score(y_test_h, rf_h_prob)

# Train XGBoost
xgb_h = XGBClassifier(eval_metric='logloss', random_state=42)
xgb_h.fit(X_train_h, y_train_h)
xgb_h_prob = xgb_h.predict_proba(X_test_h)[:, 1]
xgb_h_auc = roc_auc_score(y_test_h, xgb_h_prob)

# Train Logistic Regression
lr_h = LogisticRegression(max_iter=500, random_state=42)
lr_h.fit(X_train_h, y_train_h)
lr_h_prob = lr_h.predict_proba(X_test_h)[:, 1]
lr_h_auc = roc_auc_score(y_test_h, lr_h_prob)

print("\nHeartDisease Model Performance:")
heart_results = {
    'Heart_RF_AUC': rf_h_auc,
    'Heart_XGB_AUC': xgb_h_auc,
    'Heart_LR_AUC': lr_h_auc
}
heart_results

X_cancer = df_cancer.drop('Cancer', axis=1)
y_cancer = df_cancer['Cancer']

X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(X_cancer, y_cancer, test_size=0.2,
random_state=42)

# Train RandomForest
rf_c = RandomForestClassifier(random_state=42)
rf_c.fit(X_train_c, y_train_c)
rf_c_prob = rf_c.predict_proba(X_test_c)[:, 1]
rf_c_auc = roc_auc_score(y_test_c, rf_c_prob)

# Train XGBoost
xgb_c = XGBClassifier(eval_metric='logloss', random_state=42)
xgb_c.fit(X_train_c, y_train_c)
xgb_c_prob = xgb_c.predict_proba(X_test_c)[:, 1]
xgb_c_auc = roc_auc_score(y_test_c, xgb_c_prob)

# Train Logistic Regression
lr_c = LogisticRegression(max_iter=500, random_state=42)
lr_c.fit(X_train_c, y_train_c)
lr_c_prob = lr_c.predict_proba(X_test_c)[:, 1]
lr_c_auc = roc_auc_score(y_test_c, lr_c_prob)

print("\nCancer Model Performance:")
```

```
cancer_results = {
    'Cancer_RF_AUC': rf_c_auc,
    'Cancer_XGB_AUC': xgb_c_auc,
    'Cancer_LR_AUC': lr_c_auc
}
cancer_results


import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import roc_curve, roc_auc_score

# Step 1: Store ROC-AUC results in a structured DataFrame
results = {
    'Disease': ['Diabetes', 'Diabetes', 'Diabetes',
                'Heart Disease', 'Heart Disease', 'Heart Disease',
                'Cancer', 'Cancer', 'Cancer'],
    'Model': ['RandomForest', 'XGBoost', 'LogisticRegression'] * 3,
    'ROC_AUC': [
        0.8370, 0.8257, 0.8226,   # Diabetes
        0.9316, 0.9203, 0.9213,   # Heart Disease
        0.9972, 0.9951, 0.9928    # Cancer
    ]
}

df_results = pd.DataFrame(results)
print("=== Combined Performance Table ===")
display(df_results)

# Step 2: Bar plot for ROC-AUC comparison
plt.figure(figsize=(10,6))
sns.barplot(data=df_results, x='Disease', y='ROC_AUC', hue='Model')
plt.title("Model Performance (ROC-AUC) for All Diseases")
plt.ylim(0, 1)
plt.ylabel("ROC-AUC Score")
plt.legend(title='Model')
plt.show()

# Step 3: Function to plot ROC curve
def plot_roc(y_true, y_probs_dict, disease_name):
    plt.figure(figsize=(7,6))
    for model_name, probs in y_probs_dict.items():
        fpr, tpr, _ = roc_curve(y_true, probs)
        auc_score = roc_auc_score(y_true, probs)
        plt.plot(fpr, tpr, label=f"{model_name} (AUC={auc_score:.3f})")
    plt.plot([0,1],[0,1],'k--')
```

```
    plt.xlabel("False Positive Rate")
    plt.ylabel("True Positive Rate")
    plt.title(f"ROC Curve - {disease_name}")
    plt.legend()
    plt.show()

# Step 4: Prepare probability predictions for ROC curves
# Diabetes
rf_d_prob = rf.predict_proba(X_test_dia)[:,1]
xgb_d_prob = xgb.predict_proba(X_test_dia)[:,1]
lr_d_prob = lr.predict_proba(X_test_dia)[:,1]

# Heart Disease
rf_h_prob = rf_h.predict_proba(X_test_heart)[:,1]
xgb_h_prob = xgb_h.predict_proba(X_test_heart)[:,1]
lr_h_prob = lr_h.predict_proba(X_test_heart)[:,1]

# Cancer
rf_c_prob = rf_c.predict_proba(X_test_cancer)[:,1]
xgb_c_prob = xgb_c.predict_proba(X_test_cancer)[:,1]
lr_c_prob = lr_c.predict_proba(X_test_cancer)[:,1]

# Step 5: Plot ROC curves for each disease
plot_roc(y_test_dia, {'RandomForest': rf_d_prob, 'XGBoost': xgb_d_prob, 'LogReg': lr_d_prob},
"Diabetes")
plot_roc(y_test_heart, {'RandomForest': rf_h_prob, 'XGBoost': xgb_h_prob, 'LogReg': lr_h_prob},
"Heart Disease")
plot_roc(y_test_cancer, {'RandomForest': rf_c_prob, 'XGBoost': xgb_c_prob, 'LogReg': lr_c_prob},
"Cancer")
```

CHAPTER-V

SYSTEM TESTING

5.1 Testing

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

Types of Tests

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing:

written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

5.2 Test Strategy and Approach

Field testing will be performed manually and functional tests will be written in detail.

Test Objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features To Be Tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g., components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

5.3 Test Strategy and Approach

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

CHAPTER-VI

CONCLUSION & FUTURE ENCHANCEMENTS

6.1 Conclusion

The project "Smart Explainable AI Model for Personalized Disease Risk Assessment and Early Disease Prediction" has been able to demonstrate the implementation of AI, Machine Learning, and Explainable AI in the field of healthcare to develop a model that is capable of predicting the occurrence of certain diseases. The suggested system utilizes modern ML approaches, including Random Forest, XGBoost, and Ensemble methods, which allow predicting various diseases with a high degree of precision. Using explainability approaches such as SHAP and LIME makes the process more transparent and helps to provide patients and healthcare workers with the necessary information about how different medical factors affect the predictions made by the system.The system is capable of analyzing patients' medical, lifestyle, and demographic information to develop personalized risk assessment results and recommendations. An efficient web-based application has been developed to provide users with an opportunity to interact with the system by entering health-related information and receiving predictions.

Major accomplishments of the project:

- Accuracy of multiple disease prediction by ensemble AI algorithms
- Use of Explainable AI for clear decision making
- Personalized health risk assessment
- User-friendly website for easy interaction
- Effective management of healthcare data
- Visualization of features significance by SHAP and LIME
- Enhanced accuracy and explainability of predictions

Future Enhancements:

Several advancements can be made to the suggested system to improve its performance in practice. Some of these improvements include the addition of new advanced technologies and features that can enhance accuracy, efficiency, usability, and scalability of the system.

First, the system can be generalized to account for other types of diseases and disorders, such as kidney disorder, liver disorder, cancers, and disorders of the nervous system. Moreover, state-of-the-art deep learning methods like ANN, CNN, and RNN may be used to optimize the model's predictive performance through better prediction on large datasets.

Furthermore, the system can be developed as an app for Android and iOS operating systems in order to provide healthcare predictions at any place and time when needed. Moreover, integration of various health sensors, such as those available on wearables such as smartwatches and fitness trackers, can provide information on health parameters, such as pulse rate and blood sugar level.

Further development of Explainable AI tools and dashboards can be utilized to create detailed explanations about prediction results. Suggestions on personalized medicine including nutrition, exercise, and preventative care could also be incorporated. Better methods of ensuring privacy through enhanced security and encryption could further be included in the system. All these improvements would help make the system more intelligent and applicable in today’s world of healthcare.

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